

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 08 · Resolving Forces and Inclined Planes

Questions

Student book · questions with writing space

- 25 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 08

Resolving Forces, Inclined Planes

Your classified route

10

QUESTIONS

112

MARKS

2

FAMILIES

6

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 08 | FAMILY 01

Motion on a rough inclined plane

WHAT THIS FAMILY ASKS

Resolve forces on a moving rough-plane particle and link $F = ma$ to later kinematics.

VARIANT ROUTE

V01 Single rough-plane particle

V02 Upward and downward stages

V03 Applied force/string removed

3 classified questions

39 total marks

6.

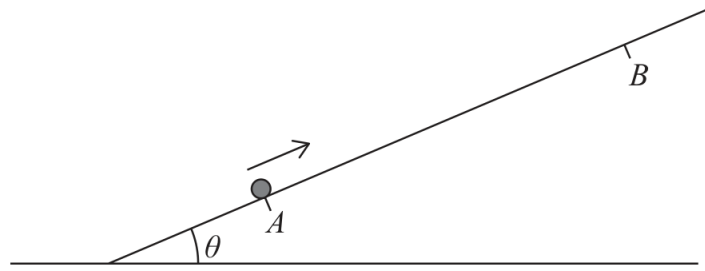


Figure 4

Figure 4 shows a rough plane inclined at an angle θ to the horizontal, where $\tan \theta = \frac{3}{4}$

The points A and B lie on a line of greatest slope of the plane, with B above A .

A package P of mass m is projected up the plane from A towards B .

The coefficient of friction between the plane and the package is $\frac{1}{4}$

The package P is modelled as a particle.

(a) Show that the **deceleration** of P , as it moves from A to B , is $\frac{4}{5}g$

(6)

Figure 4 shows a rough plane inclined at an angle θ to the horizontal, where $\tan \theta = \frac{3}{4}$

The points A and B lie on a line of greatest slope of the plane, with B above A .

A package P of mass m is projected up the plane from A towards B .

The coefficient of friction between the plane and the package is $\frac{1}{4}$

The package P is modelled as a particle.

The package P comes to rest at B .

Given that P is projected from A with speed $U \text{ m s}^{-1}$ and that $AB = 1.5 \text{ m}$,

(b) find the value of U .

(2)

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Figure 4 shows a rough plane inclined at an angle θ to the horizontal, where $\tan \theta = \frac{3}{4}$

The points A and B lie on a line of greatest slope of the plane, with B above A .

A package P of mass m is projected up the plane from A towards B .

The coefficient of friction between the plane and the package is $\frac{1}{4}$

The package P is modelled as a particle.

On reaching B , P is held at rest there by a force of magnitude X newtons acting up the plane in the direction AB .

Given that the mass of P is 2 kg,

(c) find the smallest possible value of X .

(3)

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6. A fixed rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed 5 m s^{-1} from a point A on the plane, up a line of greatest slope of the plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

- (a) Find the magnitude of the frictional force acting on the particle as it moves up the plane.

(3)

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6. A fixed rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed 5 m s^{-1} from a point A on the plane, up a line of greatest slope of the plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

The particle comes to instantaneous rest at the point B .

- (b) Find the distance AB .

(5)

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6. A fixed rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

A particle of mass 6 kg is projected with speed 5 m s^{-1} from a point A on the plane, up a line of greatest slope of the plane.

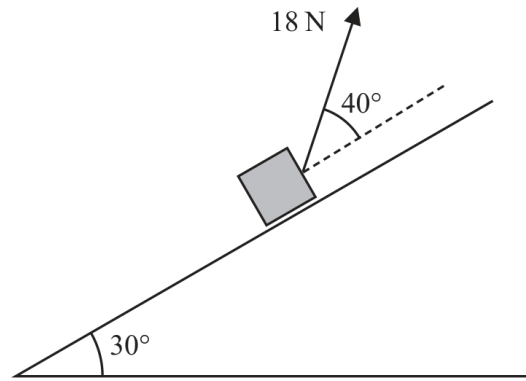
The coefficient of friction between the particle and the plane is $\frac{1}{4}$

The particle now slides down the plane from B . At the instant when the particle passes through the point C on the plane, the speed of the particle is again 5 m s^{-1}

- (c) Find the distance BC .

(5)

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A parcel of mass 2 kg is pulled up a rough inclined plane by the action of a constant force.

The force has magnitude 18 N and acts at an angle of 40° to the plane.

The line of action of the force lies in a vertical plane containing a line of greatest slope of the inclined plane.

The plane is inclined at an angle of 30° to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is 0.3

The parcel is modelled as a particle P

(a) Find the acceleration of P

(8)

A parcel of mass 2 kg is pulled up a rough inclined plane by the action of a constant force.

The plane is inclined at an angle of 30° to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is 0.3

The points A and B lie on a line of greatest slope of the plane, where $AB = 5$ m and B is above A . Particle P passes through A with speed 2 ms^{-1} in the direction AB .

(b) Find the speed of P as it passes through B .

(3)

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A parcel of mass 2 kg is pulled up a rough inclined plane by the action of a constant force.

The plane is inclined at an angle of 30° to the horizontal, as shown in Figure 5.

The coefficient of friction between the plane and the parcel is 0.3

The force of 18 N is removed at the instant P passes through B . As a result, P comes to rest at the point C .

- (c) Determine whether P will remain at rest at C . You must show all stages of your working clearly.

(4)

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TOPIC 08 | FAMILY 02

Limiting equilibrium on a rough inclined plane

WHAT THIS FAMILY ASKS

Resolve a rough-plane particle in equilibrium and apply $F \leq \mu R$ or $F = \mu R$ at a boundary.

VARIANT ROUTE

V01 Impending motion up or down the plane

V02 Two opposite rough-plane limiting states

V03 Oblique/horizontal force changes a rough-plane reaction

7 classified questions

73 total marks

8.

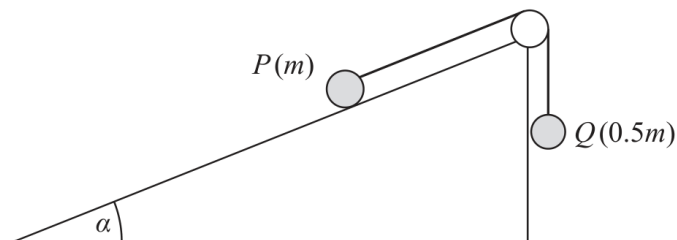


Figure 5

A fixed rough plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{5}{12}$

A small smooth pulley is fixed at the top of the plane.

One end of a light inextensible string is attached to a particle P which is at rest on the plane. The string passes over the pulley and the other end of the string is attached to a particle Q which hangs vertically below the pulley, as shown in Figure 5.

Particle P has mass m and particle Q has mass $0.5m$

The string from P to the pulley lies along a line of greatest slope of the plane.

The coefficient of friction between P and the plane is μ .

The system is in **limiting equilibrium** with the string taut and P is on the point of slipping **up** the plane.

(a) Find the value of μ .

(8)

A fixed rough plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{5}{12}$

Particle P has mass m and particle Q has mass $0.5m$

The coefficient of friction between P and the plane is μ .

The string breaks and P begins to move down the plane.

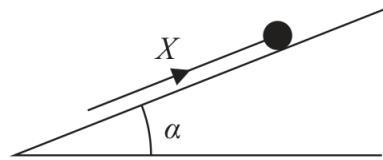
When particle P has travelled a distance of 0.8 m down the plane, the speed of P is $V \text{ m s}^{-1}$

(b) Find the value of V .

(4)

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5.

**Figure 3**

A particle of mass m rests in equilibrium on a fixed rough plane under the action of a force of magnitude X . The force acts up a line of greatest slope of the plane, as shown in Figure 3.

The plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$.

The coefficient of friction between the particle and the plane is μ .

- When $X = 2P$, the particle is on the point of sliding up the plane.
- When $X = P$, the particle is on the point of sliding down the plane.

Find the value of μ .

(8)

4.

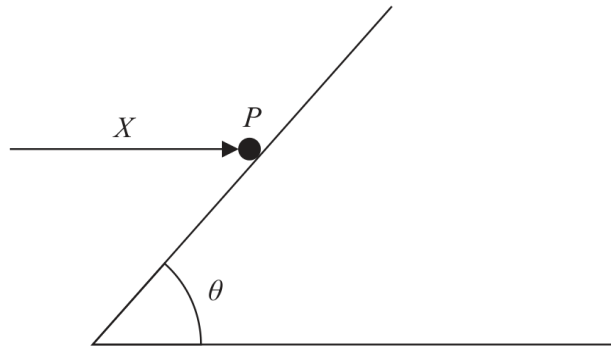


Figure 1

A particle, P , of mass km lies on a fixed rough plane. The plane is inclined to the horizontal at an acute angle θ . A horizontal force of magnitude X acts on P , as shown in Figure 1. The line of action of the force lies in the vertical plane which contains the line of greatest slope of the inclined plane that passes through P . The coefficient of friction between P and the inclined plane is μ .

When $X = mg$, the particle P is in equilibrium and on the point of sliding down the plane.

(a) Show that $\mu = \frac{k \tan \theta - 1}{k + \tan \theta}$

(10)

A particle, P , of mass km lies on a fixed rough plane. The plane is inclined to the horizontal at an acute angle θ . A horizontal force of magnitude X acts on P , as shown in Figure 1. The line of action of the force lies in the vertical plane which contains the line of greatest slope of the inclined plane that passes through P . The coefficient of friction between P and the inclined plane is μ .

When $X = mg$, the particle P is in equilibrium and on the point of sliding down the plane.

(b) Deduce that, when $k = 1$, θ must be greater than 45°

(2)

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3.

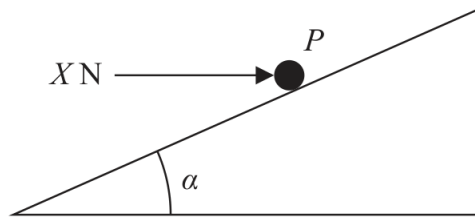


Figure 2

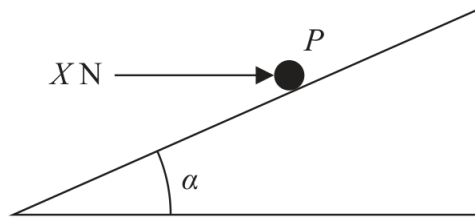
A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A particle P of mass 2 kg is held in equilibrium on the plane by a horizontal force of magnitude X newtons, as shown in Figure 2. The force acts in a vertical plane which contains a line of greatest slope of the inclined plane.

(a) Show that when $X = 14.7$ there is no frictional force acting on P

(3)

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**Figure 2**

A rough plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$

A particle P of mass 2 kg is held in equilibrium on the plane by a horizontal force of magnitude X newtons, as shown in Figure 2. The force acts in a vertical plane which contains a line of greatest slope of the inclined plane.

The coefficient of friction between P and the plane is 0.5

(b) Find the smallest possible value of X .

(8)

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6.

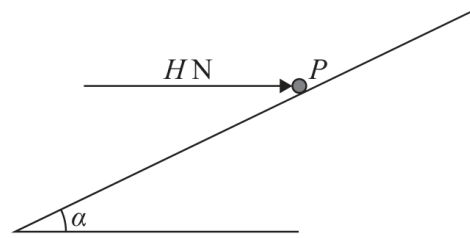


Figure 4

A particle P of mass 5 kg lies on the surface of a rough plane.

The plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$

The particle is held in equilibrium by a horizontal force of magnitude H newtons, as shown in Figure 4.

The horizontal force acts in a vertical plane containing a line of greatest slope of the inclined plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

(a) Find the smallest possible value of H .

(6)

A particle P of mass 5 kg lies on the surface of a rough plane.

The plane is inclined at an angle α to the horizontal, where $\tan \alpha = \frac{3}{4}$

The particle is held in equilibrium by a horizontal force of magnitude H newtons, as shown in Figure 4.

The horizontal force acts in a vertical plane containing a line of greatest slope of the inclined plane.

The coefficient of friction between the particle and the plane is $\frac{1}{4}$

The horizontal force is now removed, and P starts to slide down the slope.

In the first T seconds after P is released from rest, P slides 1.5 m down the slope.

(b) Find the value of T .

(6)

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6.

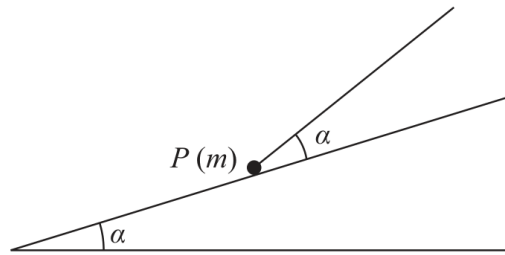


Figure 3

A particle P of mass m is held in equilibrium on a fixed rough inclined plane by a light inextensible string.

The plane is inclined at an angle α to the horizontal, where $\alpha < 45^\circ$

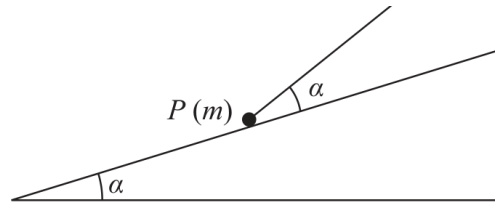
The string is inclined to the plane at angle α , as shown in Figure 3.

The string lies in a vertical plane that contains a line of greatest slope of the inclined plane.

When the tension in the string is $0.75mg$, P is on the point of moving up the plane.

- (a) Find an expression for the magnitude of the frictional force acting on P , giving your answer in terms of m , g and α .

(3)

**Figure 3**

A particle P of mass m is held in equilibrium on a fixed rough inclined plane by a light inextensible string.

The plane is inclined at an angle α to the horizontal, where $\alpha < 45^\circ$

The string is inclined to the plane at angle α , as shown in Figure 3.

The string lies in a vertical plane that contains a line of greatest slope of the inclined plane.

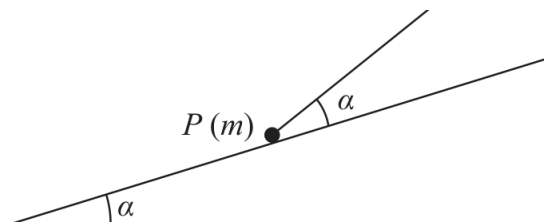
When the tension in the string is $0.75mg$, P is on the point of moving up the plane.

The coefficient of friction between P and the plane is $\frac{1}{2}$

(b) Show that

$$\tan \alpha = \frac{2}{5}$$

(6)

**Figure 3**

A particle P of mass m is held in equilibrium on a fixed rough inclined plane by a light inextensible string.

The plane is inclined at an angle α to the horizontal, where $\alpha < 45^\circ$

The string is inclined to the plane at angle α , as shown in Figure 3.

The string lies in a vertical plane that contains a line of greatest slope of the inclined plane.

The coefficient of friction between P and the plane is $\frac{1}{2}$

The string breaks.

(c) Determine whether P remains at rest. You must justify your reasoning.

(3)

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5

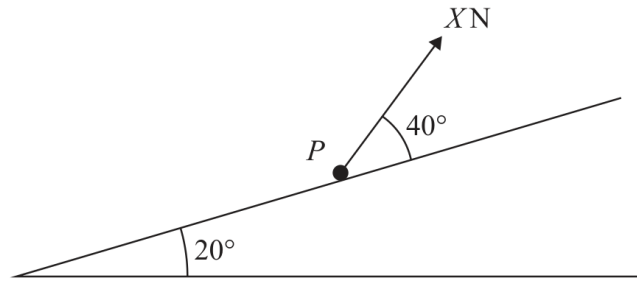


Figure 3

A particle P , whose weight is 50 N , is at rest on a rough plane. The plane is inclined to the horizontal at 20° and the coefficient of friction between P and the plane is 0.3

The particle P is held in equilibrium by a force of magnitude X newtons.

The line of action of the force makes an angle of 40° with the plane, as shown in Figure 3. The line of action of the force lies in a vertical plane which contains a line of greatest slope of the plane.

Given that P is on the point of sliding **down** the plane, find the value of X .

(6)